THE ESOTERIC LEADERS

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Innovation Project

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POLITECNICO MILANO 1863

e-novia

SOMELIA

Table of Contents

00. Executive Summary

01. The Analysis

02. The Brief

03. The Solution

04. Leadership Dynamics

05. References

00. Executive Summary

Group 36 presents *AQUA***SENSE**, an innovative solution designed for Somelia, a subsidiary of e-Novia, to revolutionize the water filtration market by integrating advanced Near Field Communication (NFC) technology and sensors.

*AQUA***SENSE** aims to enhance user experience through real-time monitoring of water filters, ensuring optimal water quality and promoting sustainability. The water filtration market is growing due to rising health consciousness and environmental concerns, yet current systems lack precise monitoring, leading to inefficiencies. Existing filters rely on estimated replacement timelines, compromising water quality and user experience.

*AQUA***SENSE** provides real-time data on filter status through NFC technology and sensors in water purifying jugs, accessed via smartphone scans for accurate filter condition updates. The proposed implementation involves collaborating with Brita to integrate *AQUA***SENSE** into their jugs, using differential pressure sensors to monitor water flow and communicate data via an NFC tag, with a mobile app offering real-time updates and predictive alerts for changing.

*AQUA***SENSE** demonstrates significant potential to transform the market through advanced technology and strategic partnerships, emphasizing continuous innovation and sustainability for long-term success for both Somelia and Brita. The report concludes with insights into the team's collaborative efforts and methodologies used to develop this innovative solution.

01. The Analysis

e-Novia 4

Somelia 5

Market Overview 6

STEEP Analysis 7

SWOT Analysis 8

Comparable Technologies 9

e-Novia

e-Novia is a deep-tech industrial group engaged in 3D design software and Product Lifecycle Management (PLM) solutions for various industries such as aerospace, automotive, machinery and electronics, providing software system services as well as technical support. The company operates by placing itself between the academy and the industry, promoting ethical and sustainable innovation, helping organisations in different sectors to turn **market opportunities into market advantages**.

Since 2015, e-Novia has been able to anticipate three trends:

- the choice and ability to operate in the field of **deep technologies**, that combine digital electronics with elements of a physical and mechanical nature, in the fields of vehicular and collaborative robotics;
- the **business creation system**, that monetizes intellectual property generated through continuous collaboration with universities and centres of research excellence, combining it with the distinctive skills of the Italian industrial fabric;
- **sustainability**, which translates into a perspective of systematic development of innovation with socio-environmental impact, placing entrepreneurship, the enhancement of human capital, the achievement of economic-financial objectives, and the adoption of transparent governance as key principles.

Today, the Group aims to serve the market with innovative products in vehicular and collaborative robotics. Its products are the outcome of an innovation process that **shapes technology into products**, designing businesses around technologies.

Somelia

Somelia, a subsidiary of the innovation-driven e-Novia enterprise, is at the forefront of integrating advanced technology within the wine industry. Founded with the mission of revolutionising how consumers interact with wine producers, Somelia develops smart solutions that enhance **transparency** and **trust** across the wine production and distribution chain.

The genesis of the company was driven by the need to address common challenges in the wine industry such as product authenticity, quality maintenance during transportation, and consumer engagement. Recognizing these industry-wide issues, Somelia sought to create a bridge between traditional wine-making practices and modern technological advancements. The enterprise's pioneering product, the **Smart Cap**, embodies this bridge by employing Near Field Communication (NFC) and specialised sensors to monitor and communicate the storage and transportation conditions of wine directly to the consumer through a user-friendly App.

This digital connection not only ensures that the integrity of the wine is maintained from the winery to the end customer but also opens up new paths for consumer interaction. By providing users with real-time data about their wine, Somelia not only protects the wine’s quality but also enriches the drinking experience, fostering a deeper understanding and appreciation of the product.

SOMELIA

Market Overview

The rising market of NFC smart tags combined with sensors is very specialised but still significant and expanding.

There is a growing embedding of IoT technology with environmental sensors for real-time data collection and analysis: this trend is particularly strong in applications like air quality monitoring and smart city initiatives, thanks to the fact that the integration of NFC technology to dedicated sensors adds value by enabling easy data access and interaction with the sensors, using compatible devices such as smartphones.

This market presents significant opportunities for **innovation**: the sensors promise enhanced features such as improved data accuracy and wider compatibility; leveraging these capabilities could lead to transformative advancements in monitoring technology, benefiting research, industry, and sustainable development initiatives.

Entering the market for NFC smart tags with sensors demands high initial investments and expertise in sensor technology, NFC, and IoT solutions. In addition the presence of key players like NXP Semiconductors leading in NFC technology and Bosch Sensortec and Honeywell dominating in environmental sensors, makes the landscape competitive.

The synergy between NFC and sensor technologies encourages partnerships and collaborations to create more integrated and efficient solutions.

STEEP Analysis

S

SOCIAL

Enhanced Experience: rising interest in understanding the status of a product in the simplest and most intuitive way.
Accessibility: higher technology inclusion thanks to simplifying the use of advanced technology.
Comfort: everyday tasks become easier.
Acceptance and Adoption: possible initial resistance due to concerns about privacy and data security.

T

TECHNOLOGICAL

Advancements: research is constantly updated.
AI & Machine Learning: integration of AI and Machine Learning enhances the capability of sensors to interpret and act on data, enabling predictive maintenance, anomaly detection and advanced analytics.
Interoperability with other technologies: ensuring interoperability with other communication technologies, enhances the versatility and utility of NFC and sensors systems**.**

E

ECONOMICAL

Market growth: the NFC sensor market is expected to grow at a CAGR of 16.7% during the forecast period from 2024 to 2032.
Cost efficiency: the initial cost of developing NFC tags and sensors is high, but the cost will gradually decrease as the technology matures and scales up production.
Investment and Funding: growing interest and investment from venture capitalists and tech giants, fostering innovation and rapid advancement.

E

ECOLOGICAL

Sustainable practices: environmental monitoring, such as tracking pollution level and waste management, contributing to sustainability efforts.
Energy efficiency: avoiding the use of batteries helps to reduce the overall environmental footprint, especially in large scale productions.

P

POLITICAL

Regulations and Standards: governments regulations and standards to ensure the safe and secure usage of the system.
Cybersecurity Legislation: with increased data collection, need for stringent cybersecurity measures and legislation to protect against data breaches and cyberattacks.
Government Initiatives: public sector investment in smart city projects and digital infrastructure can drive the adoption of these technologies, enhancing public services and urban living.

SWOT Analysis

The following SWOT analysis underscores Somelia's **strategic** positioning within a **complex** market landscape shaped by technological advancements, regulatory frameworks, and changing consumer expectations. It highlights the need for **ongoing innovation**, strategic partnerships, and proactive adaptation to regulatory environments to capitalise on opportunities and mitigate potential threats.

S

Advanced Technology: utilising NFC and environmental sensors, Somelia is at the forefront of the IoT and smart device integration, aligning with industry trends towards digitization and smart monitoring.

Consumer Engagement: the company provides technology that meets consumer demands demonstrating a commitment to enhancing user experience and satisfaction.

Expansion into New Industries: the versatility of NFC and sensor technologies allows Somelia to explore applications beyond wine, such as in other sectors where monitoring and quality assurance are critical.

W

Complexity in Integration: challenges in integrating sensor combinations into existing systems, particularly in the initial phases.

Dependence on Continuous Innovation: to remain competitive, Somelia must continuously invest in R&D to keep up with technological advances and emerging players in the market.

O

Market Growth: the market for NFC and sensor technologies is expanding, creating opportunities for businesses in various sectors.

Supportive Policies: government initiatives promoting Industry 4.0 and digital transformation.

Environmental Impact: the ecological benefits of smart monitoring technologies, align with global sustainability goals.

Industry integration: as the technology matures, it becomes more integrated and efficient, appealing to industries that require precise condition monitoring for perishable and valuable commodities.

T

Regulatory Challenges: strict data privacy and security regulations require companies to continuously adapt NFC solutions to comply with legal standards, potentially increasing costs and slowing down innovation.

Market Competition: presence of established industry leaders and the entry of new competitors with innovative solutions.

Technical Reliability: any failures or inaccuracies in sensor readings or NFC functionality could compromise the trust and reliability of incumbents, impacting customer loyalty and brand reputation.

Comparable Technologies

While analysing the innovative technology offered by Somelia, other possible technologies have been taken into consideration (*Table 1.1*).

In fact technologies like WiFi, Bluetooth, Lora and Zigbee are all part of the **wireless communication** and **connectivity** landscape; they are designed to transmit data between devices and many of them focus on low power consumption.

Even though each of these technologies has unique characteristics that make them suitable for specific applications, they all contribute to the broader goal of enabling seamless wireless communication in a connected world.

	WiFi	Bluetooth	LoRa	Zigbee	NFC+sensors
Range	Long (100 feet)	Moderate (10 metres)	Very long (rural areas)	Moderate (10-100 metres)	Very short (< 10 centimetres)
Date Rate	High	Moderate	Low	Moderate	Low
Power Consumption	High	Moderate	Low	Low	Low (zero in the sleep mode)
Privacy	Low	Mid	High	Mid	High
Discovery	Broadcast	Broadcast	Response to field	Broadcast	Response to field
Use Cases	Video streaming, remote monitoring, cloud connectivity	Wearable devices, fitness trackers, automotive connectivity, proximity-based services	Agricultural monitoring, asset tracking, environmental sensing	Smart lighting, HVAC control, asset tracking	Supply chain and logistics, smart homes, automotive, smart farming

Table 1.1: Comparison of different technologies

In the comparison, NFC combined with sensors maintains its advantages in:

- **Cost efficiency**, their components are cheap and integrate easily into existing devices, making large scale implementation affordable;
- **Speed of communication**, NFC is particularly suitable for applications where data needs to be read at regular intervals rather than continuously;
- **Low power consumption**, NFC sensors consume very little energy, making them ideal for applications where battery life and energy efficiency are critical, that’s one of the primary benefits.

02. The Brief

Selection 11

Find 13

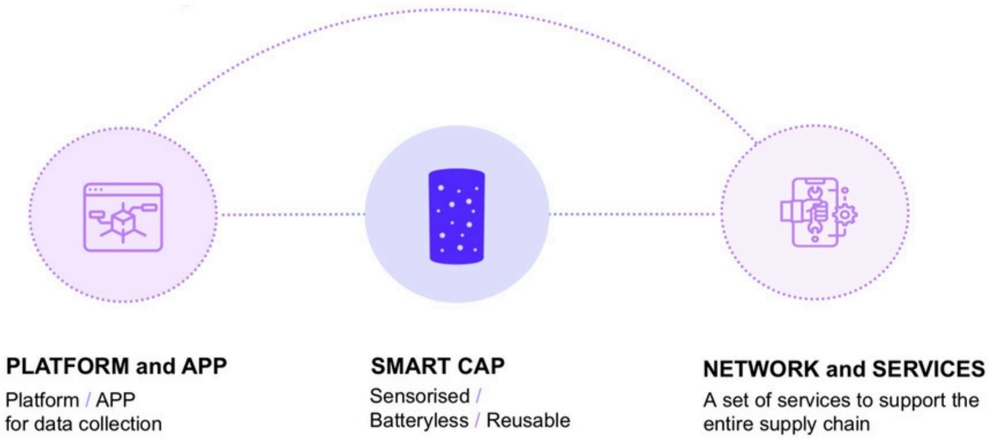
Experiment 16

Integrate 17

Selection

The intent of Somelia's solution is to **create a link** between the wine producers and customers. The idea is to involve the final consumers into the whole process, sharing information of every step the bottle went through.

In order to do so, Somelia uses an innovative technology that combines the usage of an **NFC antenna** and **environmental sensors** placed in the bottle's cap.
The sensorized Somelia capsules are reusable and battery-free, capable of detecting environmental conservation parameters such as temperature and humidity with just a touch of your smartphone.



Depending on the type of hardware configuration chosen, the Somelia Smart Cap can optionally display accurate information on bottle movements, track ageing time, and verify bottle integrity. Thanks to the dedicated App the bottle will transform into a digital interface that guides the consumer through an interactive journey to discover the wine and its story, ensuring that the bottle is stored and tasted at its best.
The technology is able to capture **instant data**: at the scan moment, the user can see the conditions and the information of the bottle of that specific time; preloaded data are available if previously scanned and saved in the proper App.

The Smart Cap is composed by three main parts.

- **NFC** (Near Field Communication) **smart tag**, a small device that uses NFC technology to transmit information or perform certain action when brought close to a compatible NFC device for less than one second in every low frequency.
- **Environmental sensors** capable of monitoring the condition of storage and the transportation in real time, collecting data about temperature, proximity, pressure, humidity and position.
- **Cork sensor** that detects small changes in wine and cork self expulsion.

Explode view

From a technical point of view this stand alone device is really intuitive and easy to use, its small size and the low power consumption characteristic (the maximum power consumption, when the tag is completely active and reading all the sensors, results to be 0,3W - 100mA at 3V - and the minimum power consumption in sensor "sleep mode", still active and capable of self-wake up but not doing anything, is 0,00003W - 10uA at 3V) make it applicable to objects in constant movement and to which battery cannot be attached for safety, environmental or reliability reasons. Furthermore this technology presents a low cost production: the reference price (lot of production > 10'000 pieces) for a tag with NFC antenna and cheap sensors is 1€.
In conclusion the technology as presented, combined with the connected platform, enables the creation of a **digital network** between stakeholders from the winery to the final consumer.

Find

Thanks to the characteristics described above, and the ability to collect various types of information, the technology proves to be highly flexible and applicable across a wide range of fields.

Nowadays, systems incorporating NFC tags and multiple sensors are utilized across various fields including supply chain and logistics, agriculture, automotive, smart homes, and healthcare. Some examples are reported below.

Supply chain and Logistics:

- Improved efficiency
- Information transparency
- Copyright assurance
- Enhanced customer experience
- Reduced costs

Agriculture:

- Soil monitoring
- Livestock management
- Water usage monitoring

Automotive:

- Keyless entry and start
- Vehicle personalization
- Vehicle maintenance and diagnostics
- Car sharing and rental
- Security

Smart homes:

- Automation
- Device control
- Security
- Convenience

Health care:

- Patient monitoring
- Smart dispensing
- Medical equipment tracking

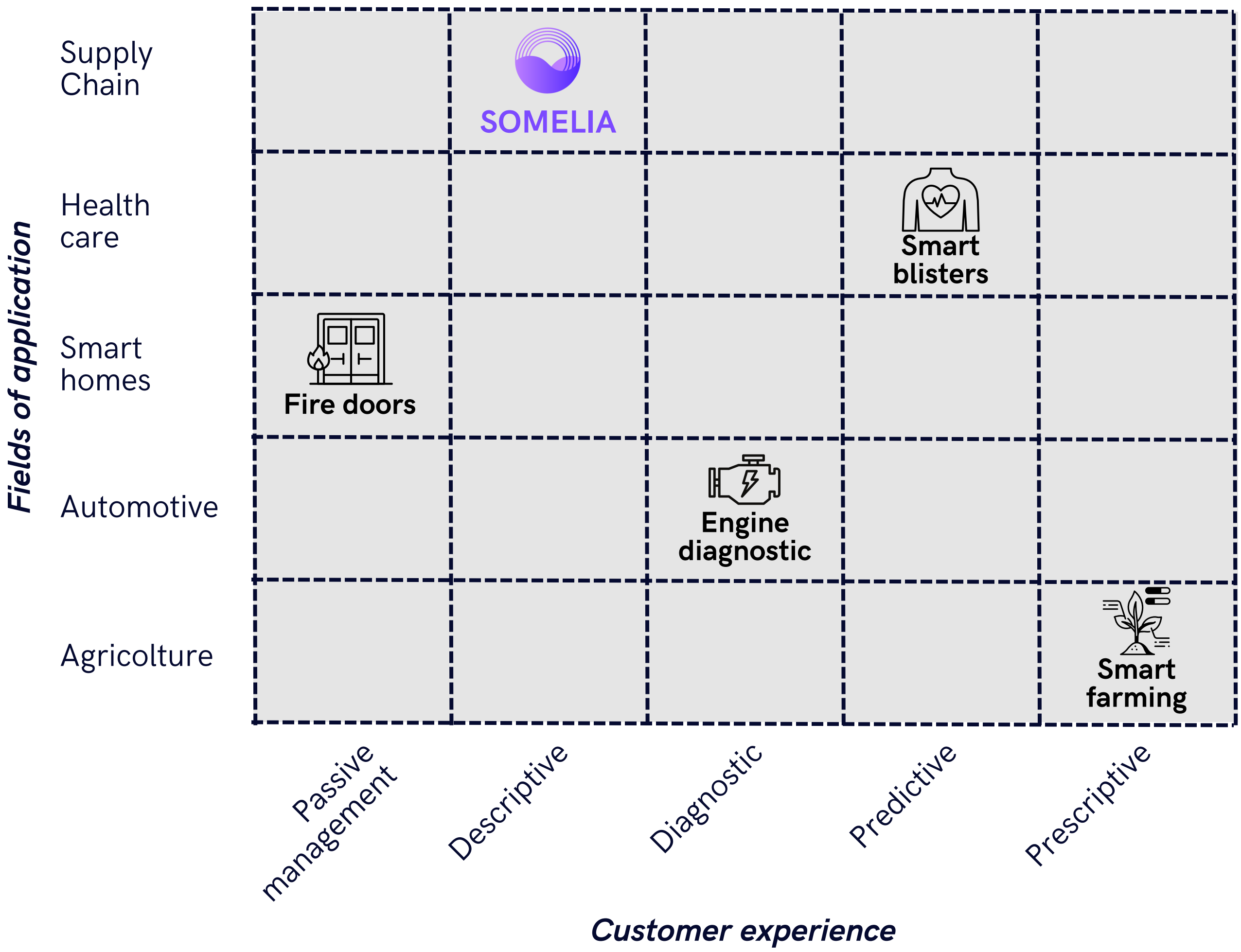
By examining existing applications, we can discern the **proposed value** to customers across various contexts. In the Smart Home domain, passive management, exemplified by fire doors, operates without direct customer engagement: temperature sensors trigger automatic door control based on preset thresholds.

Shifting focus to examples like Somelia, we enter the descriptive sphere, where user engagement reveals information actively, guiding decision-making processes.

In the automotive sector, diagnostic experiences take precedence to answer questions like "*why this event happened?*". Integrated systems identify root causes behind events, enhancing reliability and performance, offering users peace of mind.

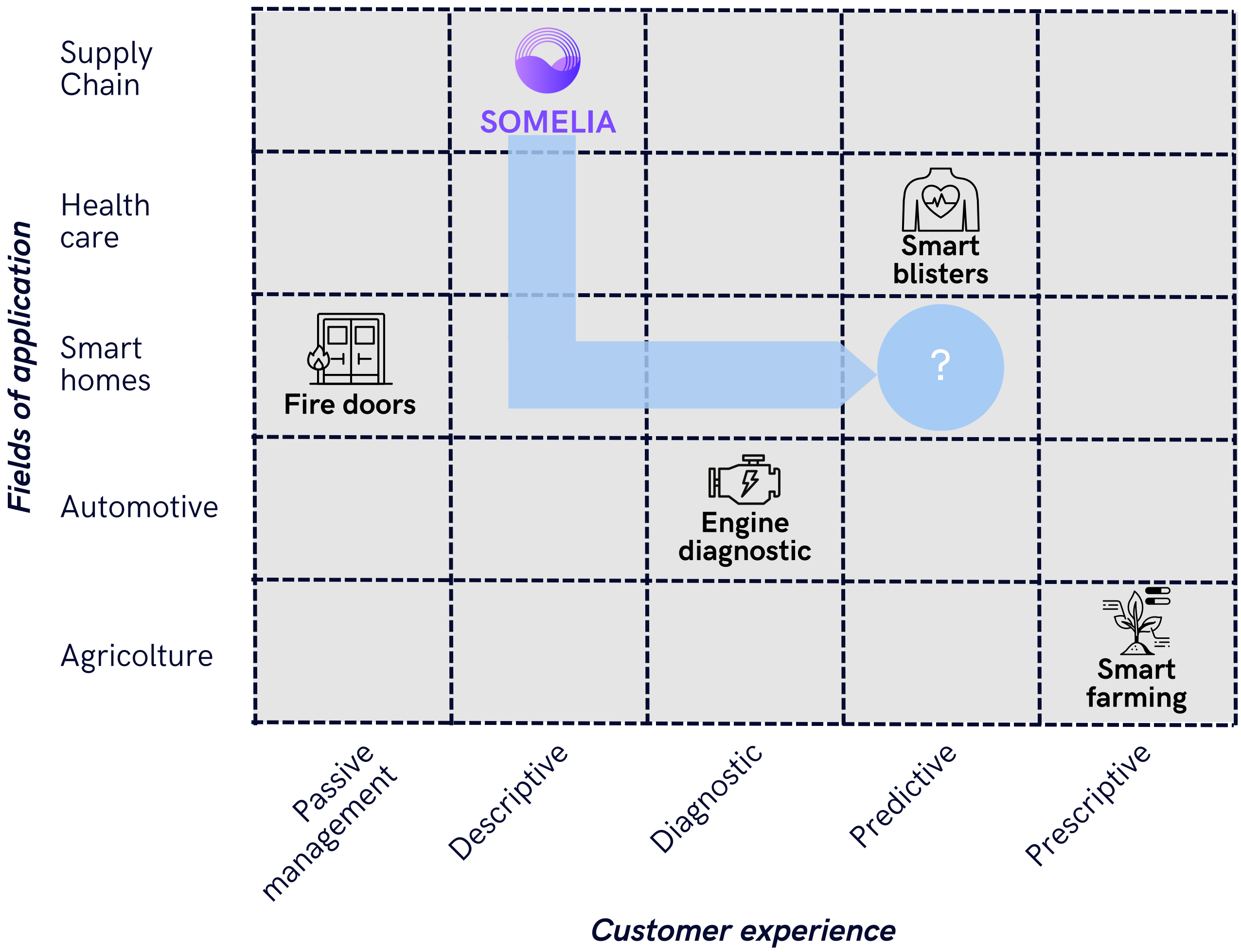
In pharmaceuticals, predictive technology comes into play. For instance, smart blister packs predict health risks by monitoring medication adherence, enabling proactive interventions to enhance outcomes.

Lastly, in agriculture, customer experiences reach a prescriptive level. Systems not only predict but also recommend actions to optimize crop health and productivity, maximizing resource efficiency.



Given the strategic focus on the smart homes sector, we aim to enhance the user experience by expanding capabilities from descriptive to diagnostic, predictive, and prescriptive levels. This approach not only allows users to ascertain the current status of items like a bottle of wine through the Somelia system but also provides a comprehensive analysis and actionable insights. Users receive detailed feedback on what is currently happening and why (diagnostic), insights into what might happen in the future (predictive), and recommendations on actions to either mitigate or leverage these outcomes (prescriptive).

This sophisticated level of interaction ensures that our smart home technology is both user-friendly and minimally invasive, seamlessly integrating into daily life while providing enhanced control and understanding of the home environment.



Experiment

An initial idea was to implement a monitoring system capable of overseeing the condition of the boiler within the home. The domestic boiler is, in fact, the only household appliance whose annual maintenance is mandatory on a yearly basis in many countries and regions. This regulation is in place to ensure the safety of heating systems and to prevent potential accidents resulting from malfunctions. Nevertheless the possibility of having other failures in the usage during the year still exists and additional maintenance is needed whenever a malfunction occurs.

What initially appeared to be strong points in favour of this idea - such as its simplicity and ease of implementation - revealed several critical issues upon closer examination.

Firstly, the technological limitations inherent in Somelia's solution significantly hindered its ability to provide customers with a truly valuable product. These constraints prevented the product from meeting customer expectations and fully delivering on its intended benefits.

Secondly, the advantage of the low production and implementation cost of this technology becomes less significant when compared to the considerable price of the boiler. Justifying a higher cost could allow for the use of more precise sensors and the implementation of a fully automated data collection system, eliminating the need for active user participation. This would relieve the user of any tasks and significantly enhance the overall user experience.

The subsequent team discussions resulted in the dismissal of that idea and the generation of new ones. Among these, one concept in particular stood out: *AQUA***SENSE**.

Integrate

Introducing *AQUA*SENSE, an innovative system designed to revolutionise your home water filtration. At the heart of this solution is a differential pressure sensor embedded within your carafe, seamlessly connected to an NFC tag. This cutting-edge technology monitors the effectiveness of your water filter.

With just a quick scan of the NFC tag using your smartphone and our user-friendly app, you gain instant access to information about the filter's status. But that's not all - the app harnesses the power of a predictive algorithm to provide personalised advice, ensuring you always enjoy the purest water possible. Say goodbye to guesswork and hello to effortless, smart water filtration with *AQUA*SENSE.

*AQUA*SENSE

Your Water, Your Way

It's common knowledge that in recent years citizens are more and more involved in **sustainability** matters: reusable bottles and carafes made their way replacing plastic bottles to reduce waste (the use of one reusable bottle can save an average of 156 plastic ones per year).

Furthermore, after COVID-19 Pandemic, people's **attention to health** has increased significantly: in 2023, an average of around 71% of people interviewed from 31 countries around the world reported they are concerned about their own physical health.

Thanks to these two trends, the market of water filters for carafes had a significant boost, solving the trade-off between the usage of reusable bottles to auto-fill at home and the concern for clean and healthy water.

This groundbreaking solution empowers users to confidently ensure the water brightness, while also reducing waste.

*AQUA*SENSE redefines the water jug, transforming it into a personal assistant that guarantees every drop is as pure as it should be. Experience a new level of convenience and assurance with smart, sustainable water filtration.

03. The Solution

AQUASENSE 19

Innovation of Meaning 21

Customer Perspective 22

Technological Perspective 25

Economical Perspective 28

Environmental Perspective 34

Business Model Canvas 36

*AQUA*SENSE

The innovative approach leverages NFC tag technology combined with advanced sensors to **enhance** the functionality of water purifying jugs. By integrating these technologies, the aim is to provide a more precise and user-friendly solution for monitoring and maintaining water filters, ensuring that the water quality remains optimal for every household.

Traditional water purifying jugs rely on general estimates to determine when a filter needs to be replaced. These estimates are usually based on average usage, which can vary significantly between different households: for instance, a family of four will naturally consume more water than a single individual, leading to a more rapid depletion of the filter's effectiveness. *AQUA***SENSE** addresses this discrepancy by offering a **personalised** approach to filter monitoring.

The system incorporates sensors that analyse the water flow during the filtration. This data is then communicated via an NFC tag embedded in the jug; when users scan the NFC tag with their smartphones, or other NFC-enabled devices, they receive real-time information about the filter's status.

This user-friendly interface provides clear notifications about when it's time to replace the filter, based on actual usage rather than a generic timeline. Additionally, by simply updating the app when the filter is changed, clients can receive a personalised estimate of the filter's remaining lifespan based on their specific usage. They receive timely notifications when the filter is nearing saturation, ensuring it remains effective and never compromised.

One of the key advantages of *AQUA***SENSE** is its seamless integration into the water purifying jug. Developed in collaboration with a leading pitcher and filter company, the device is built into the carafe at the time of manufacture, ensuring a sleek and unobtrusive design. This integration means that users do not need to worry about installing external devices or dealing with cumbersome attachments. Everything is designed to work straight out of the box, enhancing the **user experience**.

By providing accurate, usage-based information, the system not only ensures that water filters are replaced at the right time but also promotes better health and safety. Clean water is essential for maintaining good health, and this technology helps to guarantee that every glass of water from the jug meets the highest standards of purity.

In addition to improving the user experience, the presented innovation also has environmental benefits: by accurately tracking filter usage, it can help reduce unnecessary waste, filters will be replaced only when truly needed, rather than on a fixed schedule, which can lead to premature disposal of still-effective filters.

Innovation of Meaning

The innovation of meaning in the *AQUA***SENSE** lies in its ability to **transcend the conventional** approach of filter management based on average usage.

The system reinterprets this process, giving it new meaning through personalization and precision. This redefined approach, not only optimises performance and user satisfaction, but also fosters a deeper connection between the product and the consumer: the jug is no longer just a container but an intelligent partner in health and wellness, embodying a significant advancement in the meaning and value of home water purification.

from

A ***DROP*** OF CURIOSITY

to

AN ***OCEAN*** OF KNOWLEDGE

From

With Somelia, the consumer can check certain points in the wine supply chain by scanning the NFC on the bottle cap. This represents a limited interaction, in which the user only satisfies a superficial curiosity.

Consumer interaction is limited and passive. Scanning the NFC on the bottle cap provides only basic information on the touch points in the wine supply chain. This functionality represents a 'drop' of information that can only stimulate a momentary curiosity.

To

With *AQUA***SENSE** the consumer can monitor the condition of the filter in the carafe and know exactly when it is time to change it. This allows the user to have active and conscious control, accessing a greater amount of relevant and useful information.

With this new functionality, the consumer is able to access an 'ocean' of information, thus understanding and improving the efficiency and effectiveness of the carafe usage.

Customer perspective

Customers are **committed** to using the *AQUA***SENSE** and actively participate in the process of scanning the NFC sensor for several compelling reasons.

Health Assurance: Users are motivated by the assurance that they are drinking the cleanest and safest water possible. By scanning the NFC sensor, they can monitor the filter's status in real-time and ensure timely replacements, thus avoiding the health risks associated with expired or overused filters.

Cost Efficiency: Scanning the NFC sensor allows users to replace filters based on actual usage rather than arbitrary timelines. This means they only spend money on new filters when necessary, making the process more cost-effective and preventing unnecessary expenditures.

Convenience: The ease of use provided by NFC technology simplifies the maintenance of the water purifying jug. Customers appreciate the convenience of instantly accessing filter status information with a simple scan, which saves them time and effort compared to traditional methods.

Environmental Responsibility: Eco-conscious customers are committed to reducing waste. By using the NFC sensor to determine the exact moment a filter needs replacement, they avoid premature disposal of filters, contributing to environmental sustainability.

Peace of Mind: Regularly scanning the NFC sensor provides users with peace of mind, knowing that they are proactively managing their water quality. This reduces anxiety about the potential presence of contaminants and ensures their family's well-being.

These reasons highlight **why** customers are not only committed to using this innovative solution but also actively engage in the process of scanning the NFC sensor to ensure optimal performance and benefits.

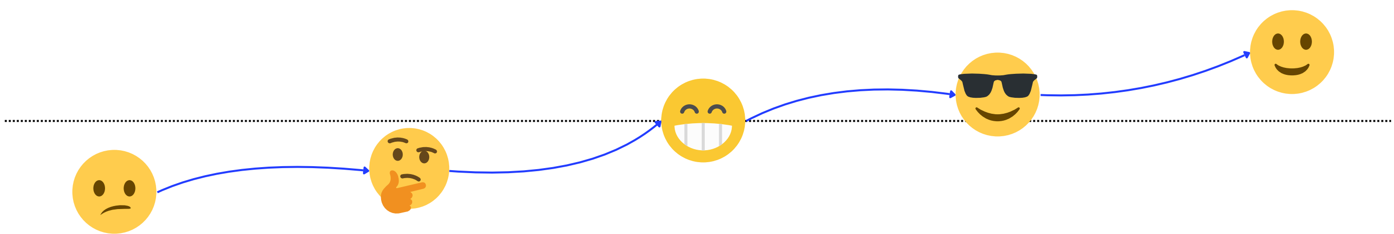
In addition, the immediate feedback received reinforces the habit of regular monitoring. Users see the direct benefits of their actions in terms of water quality and filter longevity, encouraging them to continue participating in the process actively.

Customer journey

This customer journey map provides a comprehensive overview of the **user experience** and highlights the various touchpoints, emotions, and thoughts that shape the customer's interaction with the product, ultimately leading to long-term satisfaction and loyalty.

	Awareness	Consideration	Decision	Use phase	Loyalty
User actions	The user discovers the need to improve the quality of the water he/she drinks.	The user starts researching different types of filter carafes.	The user chooses and buys a filtering carafe.	The user starts using the filter carafe, installs the filter and periodically scans it. When it is time to change the filter, the user receives an allert on his/her smartphone.	The user is satisfied and continues to use the filter carafe, possibly recommending it to others.
Touchpoints	Online advertisements, product reviews, recommendations from friends/family, articles on health and wellness.	Review websites, unboxing and review videos on YouTube, product comparisons on e-commerce.	E-commerce site, physical shop, shopping app.	Carafe, integrated NFC sensor, filter monitoring app, user manual.	Loyalty programmes, satisfaction surveys, online reviews, social media.

Experience



Emotions & Thoughts	Curiosity, health concern	Interest, uncertainty	Excitement, relief	Satisfaction, practicality, mind peace	Loyalty, pride, satisfaction
	"The water in my house tastes strange" *"I've heard that filtered water is healthier"*	*"What is the best filter carafe on the market?"* *"What are the important features?"*	*"This model seems to have good reviews"* *"It has all the features I need"*	*"The water really does taste better"* *I DO NOT have to remember to change the filter regularly"*	*"This product is great, I will recommend it"* *"I can no longer do without my filter carafe"*

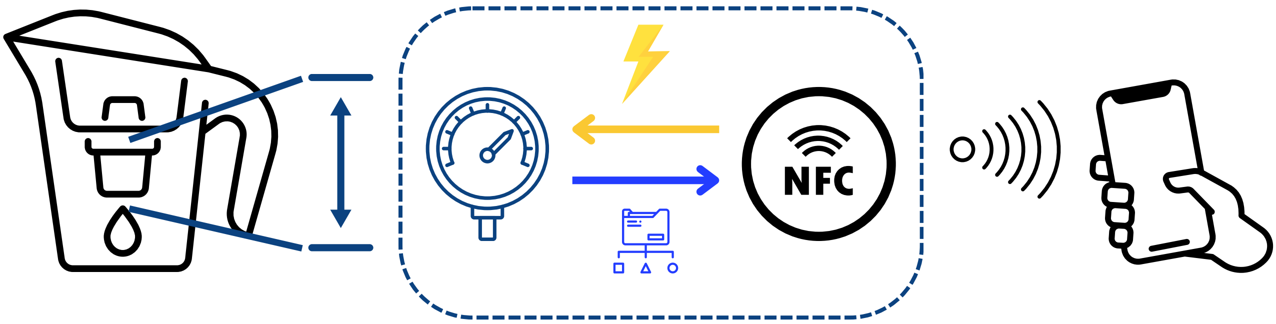
Technological Perspective

The sensors

To accurately monitor the effectiveness of water filters, the system employs a **differential pressure sensor**. This sensor features two pressure ports placed strategically before and after the filter. By measuring the inlet and outlet pressures, it can calculate the pressure difference. Based on this differential pressure and pre-determined laboratory values and assumptions, the system can determine whether the filter remains usable. This information is then conveyed to the user via an NFC tag.

This choice of differential pressure to measure flow rate, rather than other parameters, is carefully considered: flow rate and the saturation of activated carbons are the only parameters that can reliably indicate filter effectiveness without requiring memory or a battery; activated carbons, which are widely used in water filters, absorb impurities through their micro-pores, once saturation is reached, they stop functioning effectively.

However, measuring carbon saturation directly is only feasible in a laboratory setting, therefore, differential pressure has been used as a practical proxy. As the filter saturates, it impedes water flow, increasing the pressure difference and, by monitoring this difference, the system provides an accurate and real-time assessment of filter status.

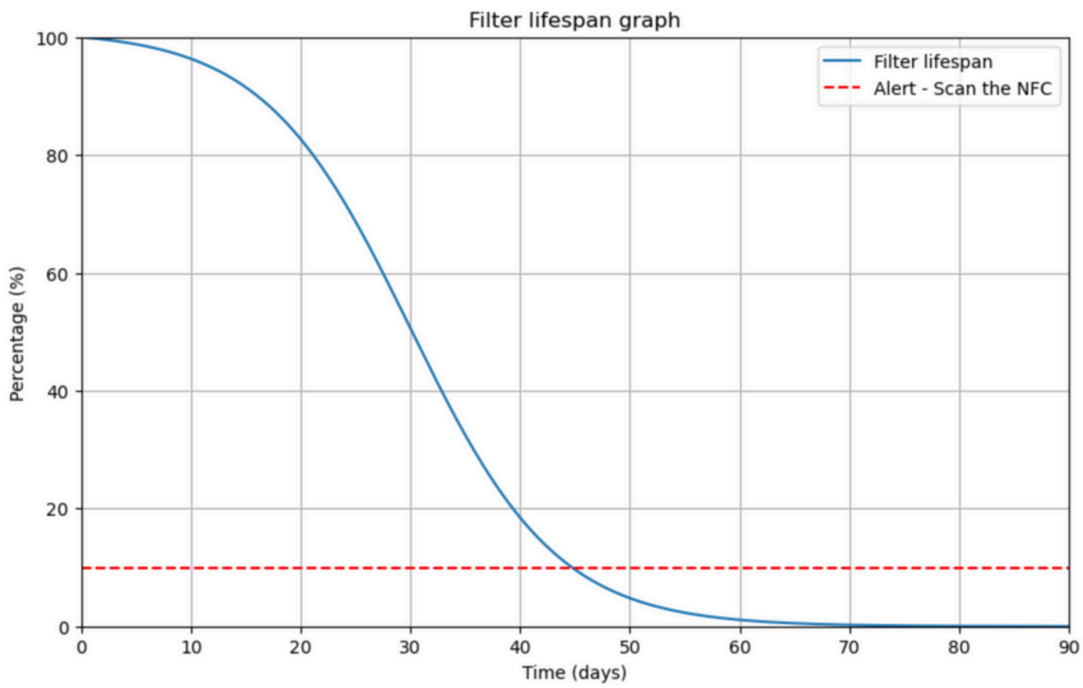


AQUASENSE

The study behind

The measured values will have to be compared with predetermined values in the laboratory. In order to construct an adequate control system it will be necessary to make assumptions regarding the quality of the incoming water. In fact, activated carbon filters mainly retain chlorine, organic compounds and heavy metals, obviously the greater the amount of these compounds the higher the saturation rate of the filter will be, and therefore, once full, the lower the final outflow rate due to the resistance of the materials with the filter. To this, however, it can be assumed that the water in general is taken from the home tap, and therefore it will be sufficient to test with water that falls within the average values of the above components.

Once the water to be used has been chosen, the experiments with the filters will be carried out. A sufficiently large sample of measurements will need to be created, in which for each filter the water flow rate throughout its life cycle will be evaluated; once the data sample is obtained, the trend is studied, which will be negative and directly proportional to the saturation of the filter, stabilising when it begins to stop working. At the end, a control system can be constructed that will indicate the threshold limit within which the filter is still usable.



On the other hand, the **predictive** algorithm that allows for an estimate of the filter's lifetime, it is simple to design but has a few hurdles within it. Once initialised by the client notifying the app that a new filter has been inserted, it saves in memory the various scans on that particular filter and calculates the average time of use, obviously the more scans the more accurate the prediction.

However, one problem turns out: assuming the client goes away for work for two months and the filter remains the same, when he returns the data that will be saved in memory by the algorithm will be completely off-kilter compared to the others, in statistics this is called an outlier. Since the data sample size is small, the outlier will have a strong influence by completely throwing off the results. For this reason it will have to be detected by the algorithm and removed. This can be done by using a Z-Score Method, which tells you how many standard deviations a data point is from the mean. Data points with Z-scores greater than a certain threshold (usually 3) can be considered outliers.

The interface with the customer

When the customer is curious to find out if it is time to change the filter, all he needs to do is open the App in which a very **user-friendly interface** will allow him to bring his phone close to the NFC tag, located on the lid of the carafe and receive a real-time approximation of the filter saturation reached, and in case alert if it is time to change it.

Finally, the app can also be predictive. In fact, if the client would like to have information not only about the status of the current filter, but also an estimate of how much longer it will last, it has the sole task of alerting the app at the time it has changed the filter. In this way, through a simple predictive algorithm it will be able to estimate the end of life of the filter based on the previous filters' lifespan of that specific client, and no longer on general estimates. This system is very effective, indeed it does not affect the quality of the measurement in any way: in case the customer forgets to enter the date of filter change, this will not alter the estimate on its real-time state, and the customer can simply start it at the next change.

Economical Perspective

In order to enter and to compete effectively in the market it has been explored the trends in hydration choices around the world.

Plastic Bottle

Plastic bottle consumption has been a significant concern due to its environmental impact. However, there has been a noticeable shift towards reducing plastic usage in recent years. Many consumers are becoming more conscious of the environmental implications and are actively seeking alternatives.

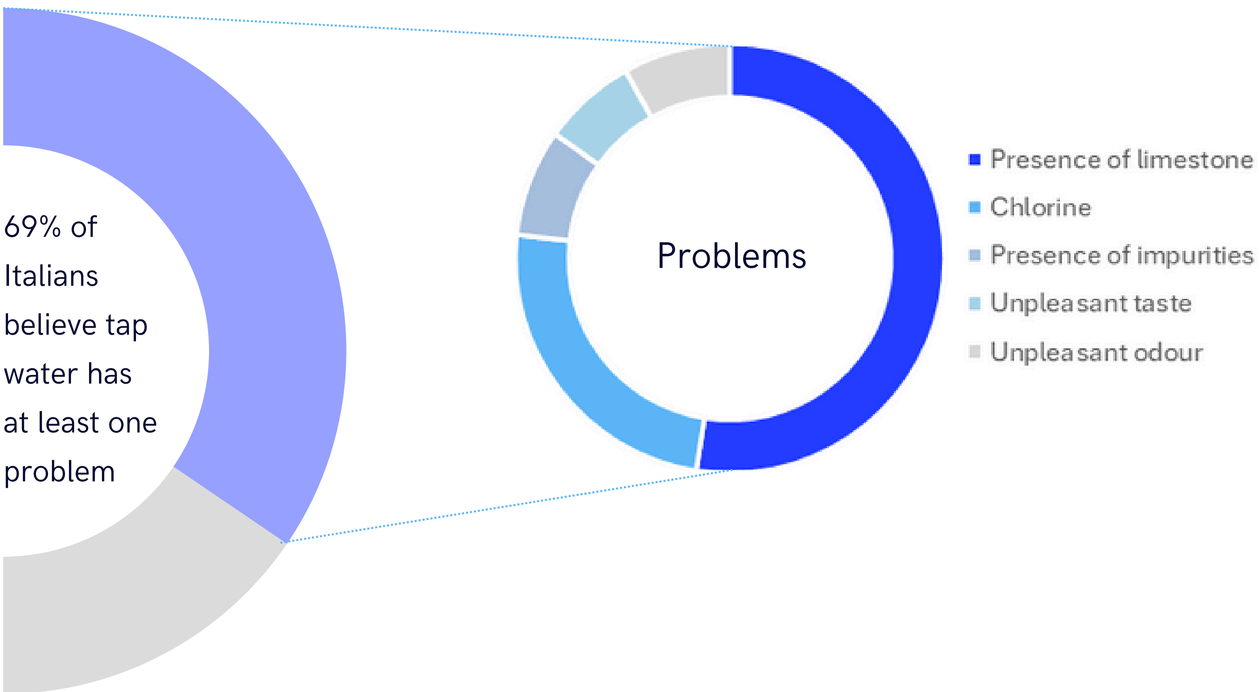
Reusable Glass Bottle

Reusable glass bottles have gained popularity as a more sustainable option compared to single-use plastic bottles. They are often perceived as safer for consumption and more environmentally friendly. Many companies and brands have started offering water in reusable glass bottles to cater to this demand.

Tap filtered water

Filtered water from the tap, served in carafes, has also seen an increase in popularity. This option provides consumers with a convenient and cost-effective way to enjoy clean and safe drinking water without the need for single-use bottles. It aligns with the growing trend towards sustainability and reducing plastic waste.

Tap water's quality is positively considered by 76% of Italians yet only 28% of them consume it, filtered, constantly. Interesting data if compared to other European countries such as France, where the percentage is up to 46% and Spain with a peak of 55%.
69% of respondents believe that tap water has at least one problem. First of all, water hardness: excessive presence of limestone is indeed cited as a problem by 52% of Italians. In second place, with 24% of responses, is the presence of chlorine. Followed in general by the presence of impurities at 8%, unpleasant taste cited by 7% of respondents, and unpleasant odour for 8%.
In recent years there is a growing awareness and preference for sustainable options among Italian consumers. This is driving the demand for alternatives to plastic bottles, such as reusable glass bottles and filtered tap water. Furthermore clients are also becoming more health-conscious preferring water options that are perceived safer and cleaner. This point of view has changed over time since 20 years ago when bottle waters were considered more pure than tap ones. In addition customers are taking into account the economic convenience that plays a significant role, indeed companies that offer convenient and eco-friendly solutions are likely to succeed in the market.



Players in the water purifies market

The technology must be implemented in a pitcher in order to provide a finite product for consumers. Because of this, an analysis was conducted to detect the **principal actors** in the aimed market, to select the most suitable partner.
Carrying meticulous research, the following European players have been identified as producers of water purifiers.

Brita: a well-known brand offering various types of water filtration products, including pitchers, faucet filters, and filter cartridges.

PUR: another prominent player in the filtered water market, offering a range of filtration products for both home and commercial use.

Aqua Optima: this company specialises in water filtration products, including filter jugs, filters, and filter bottles, with a focus on sustainability.

BWT: Best Water Technology (BWT) is an European company providing water treatment solutions, including filtration systems for domestic and commercial use.

ZeroWater: ZeroWater is known for its advanced filtration technology that claims to remove virtually all dissolved solids from tap water.

TAPP Water: offers eco-friendly water filtration solutions, including faucet filters and biodegradable filter cartridges.

Bobble: Bobble is recognized for its stylish and portable filtered water bottles, offering on-the-go filtration for consumers.

Positioning

Brita has been identified as the foremost player in the European water purifier market, boasting a robust turnover of €664 million in 2022 and expanding globally, making them a potential partner. In comparison to competitors in the water filtration field, Brita holds a strong market position with a well-established brand image and a reputation for quality and reliability. Given its market leadership, Brita presents an opportunity that could mutually benefit both parties.

BRITA

X

AQUASENSE

AQUASENSE

Cost analysis

In the analysis, the possibility of making the NFC tag **reusable** across different filter cycles, rather than a single-use item, have been explored. While attaching it to the carafe poses logistical challenges for partnering with Brita, integrating the NFC tag with the filter itself could be more complex. The latter alternative would entail disposing of the NFC tag with each used filter, potentially leading to more frequent purchases, increased revenue, as well as waste and environmental impact.

Brita's filters are currently sold at an average price of €6 per unit, but Brita is not the only contender: other companies, such as EverSpring, have entered the market producing filters suitable for Brita's jug, offering a lower price. On the other hand, concerning jug prices, they can be purchased starting from less than €20; once bought, they should last for years, so the company's revenues in this scenario come at once. All these factors have been considered to make the right choice regarding a potential partnership. After reviewing Brita's jug catalogue, the most intriguing scenario is to develop a **new carafe**, produced in collaboration with Brita's design team, embedding the presented technology. Brita currently offers six types of carafes with a minimum price of €18.99 for the plastic model and a maximum price of €59.99 for the premium model.

The new product, named '**BRITA** x *AQUA***SENSE**', will be made of plastic, aligning with most of Brita's product line. It will be positioned in the market with a price targeting the average consumer who prioritises health and environmental impact, utilising the predictive features brought by the overall system.

The concept involves integrating sensors near the filter's entrance and exit, connecting it with a cable built-in the carafe, to the NFC system. This one will replace the small interactive display currently available in the market, as illustrated in the figure.

Unlike the current display, which has an integrated battery lasting around four years before requiring replacement (at a cost of €10), this won't be necessary due to the long-lasting nature of the NFC tag.

Producing a pressure-detecting sensor, essential for the solution's requirements, on average costs less than €2.5. Additionally, the NFC technology required comes at a cost of around 20 cents to which is added the cost of the wire connection. Finally, a 20% markup is added. This markup level is justified by the average markup applied to innovative products in the smart home sector (Table 3.1).

Taking as a reference Somelia’s manager indications, these costs can be achieved through economies of scale when the production volume is set around 10'000 pieces.

Among various pricing strategies, the cost-plus markup pricing method has been chosen to remain competitive in the market, facilitating better market penetration.

Due to its modern design and plastic construction, the “Style” carafe was selected from Brita’s catalogue as a reference product in the targeted market. Starting with the market price of the carafe, the cost of the embedded technology is added (Table 3.2).

Description	**Cost [€]**
NFC tag + 2 pressure sensor	5.20
Integrated copper cable (15 cm)	0.10
Total cost	**5.30**
+ Markup	+ 20%
Cost + markup	**6.36**

Table 3.1: Total costs of the technology

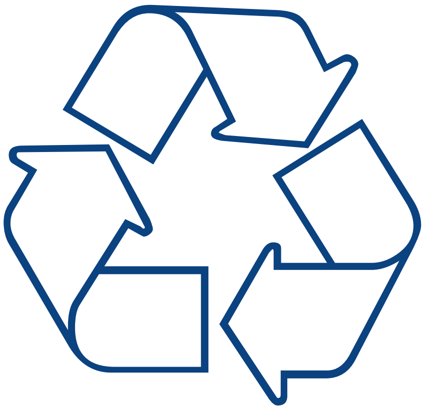
Description	**Price [€]**
"Style” carafe price	31.99
Total cost + markup	6.36
TOTAL	**38.35**

Table 3.2: Selling price definition

Environmental Perspective

Nowadays the filters used for carafes have a lifespan of 100/150 litres, but it's unfeasible with the current devices, to know when it's time to get a new one. To address this issue, companies assume an **average** consumption of 3.5 litres per day. With this value, it's possible to advise customers to change their filters once a month and, to help users keep track of this, carafes are equipped with a small display that counts 4 weeks.
However, this method has several drawbacks in terms of user experience: customers must remember to initiate the timer, which frequently deteriorates over time due to its low quality.
Consequently, consumers often lack awareness of when to replace the filter, leading to premature replacements or, on the contrary, continued use beyond its effectiveness due to forgetfulness.

Focusing on the filters, they consist of plastic wrapping, activated carbon and an ion exchanger, and each component can be recycled through the following steps.
The plastic wrapping is initially crushed on-site and then collected by partner companies, who clean and separate the plastic, grinding it into pellets, which can then be used by the plastic industry to produce new plastic parts.
The activated carbon is returned to its suppliers, where it’s refined and reused for various filtering processes, such as wastewater treatment.



Finally, the ion exchanger is regenerated and added to new filter cartridges: this means that the new BRITA filter cartridges can contain regenerated ion exchangers, maintaining the same quality and performance.

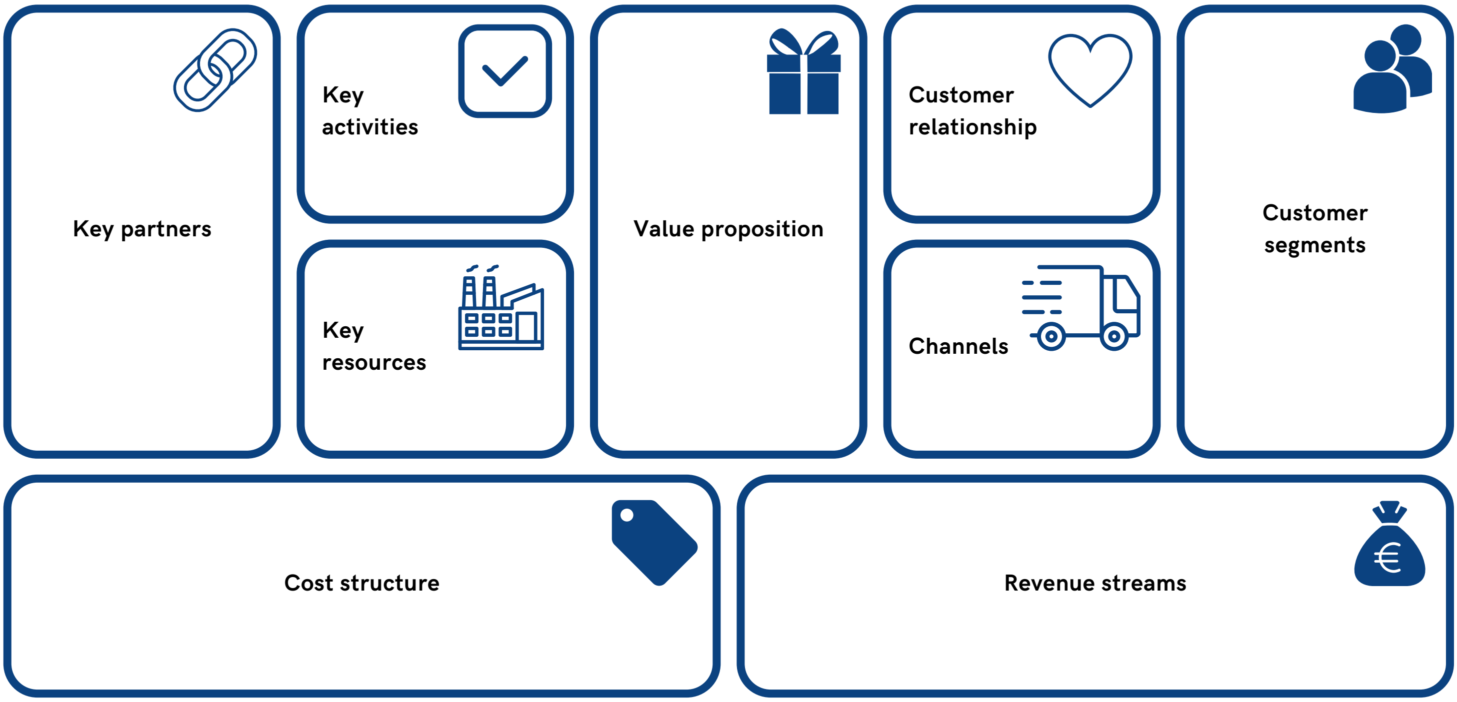
Following a **Triple Bottom Line** model, the first step is the efficient use of technologies and, since filters are completely recyclable, it's important to collect used filters and know when they are finished, to reduce the number of filters discarded before reaching their lifespan. Now with *AQUA***SENSE**, it’s possible to determine that moment precisely and additionally, with the app, users can be reminded that filters should not be thrown in normal waste but collected by the company.

The advantages don’t end here; the second step is to be eco-friendly. Therefore, it has been devised a way to conserve NFC usage: instead of embedding NFC in the filter, it will be incorporated into a new type of carafe equipped with this technology. This approach eliminates the need to collect NFC, allowing it to be used for as long as the carafe is in circulation, reducing the number of units produced and consequently minimising the environmental impact.

In this way, innovation can be seen as a tool to improve sustainability. In fact, the solution transforms an existing product into an instrument to reduce environmental impact as much as possible and use resources efficiently, but also enhancing social well being by improving the quality of water as the third and last step of the Triple Bottom Line.

Business Model Canvas

The Business Model Canvas outlines a comprehensive approach to developing, marketing, and selling a technological solution that enhances traditional water purifying jugs with NFC and sensor technology. The focus is on delivering **personalised**, **user-friendly**, and **environmentally sustainable** products that meet the evolving needs of health-conscious and ecologically aware consumers. By leveraging partnerships with established brands like Brita and focusing on innovation and customer experience, this business model aims to create significant value for both consumers and stakeholders.



POLITECNICO MILANO 1863

Water Filter Manufacturers: Partnership with leading companies like Brita to integrate the technology into their products.
Technology Providers: Suppliers of NFC tags and sensors.
Research Institutions: Collaboration for testing and validating the effectiveness of the sensors and filters.
Recycling Companies: To handle the recycling of used filters.

Product Development: Designing and integrating sensors and NFC technology into water purifying jugs.
Quality Assurance: Ensuring the accuracy and reliability of the sensor data.
Marketing and Sales: Promoting the product through various channels to reach the target market.

Technology: NFC tags, differential pressure sensors, mobile app development.
Intellectual Property: Patents for the integrated technology.
Human Resources: Engineers, researchers, marketing, and sales teams.

Personalised Filter Monitoring: Accurate, instant data on filter status based on actual usage rather than estimates.
User-Friendly Experience: Easy-to-use NFC scanning for instant information on filter condition.
Health and Safety: Ensures optimal water quality by timely filter replacements.
Environmental Benefits: Reduces unnecessary waste by replacing filters only when needed.

Direct Interaction: Through the mobile app that provides real-time updates and notifications.
Community Building: Online forums and social media for sharing tips.

Retail Stores: Physical stores selling water purifying jugs and filters.
Online Sales: E-commerce platforms and the company's website.
Mobile App: Provides updates, notifications, and user guides.

Health-Conscious Individuals: Users who prioritise clean and safe drinking water.
Environmentally Conscious Consumers: Users looking to reduce plastic waste and promote sustainability.
Tech Enthusiasts: Early adopters of innovative smart home products.

R&D Costs: Research and development of sensors and NFC integration.
Marketing and Sales Costs: Advertising, promotions, and sales commissions.
Technology Costs: Costs associated with maintaining the mobile app and data services.
Partnership Costs: Expenses related to the collaboration with Brita.

Product Sales: Revenue from selling the BRITA x *AQUA*SENSE jugs.
Data Analytics: Offering anonymized usage data for research and market analysis.

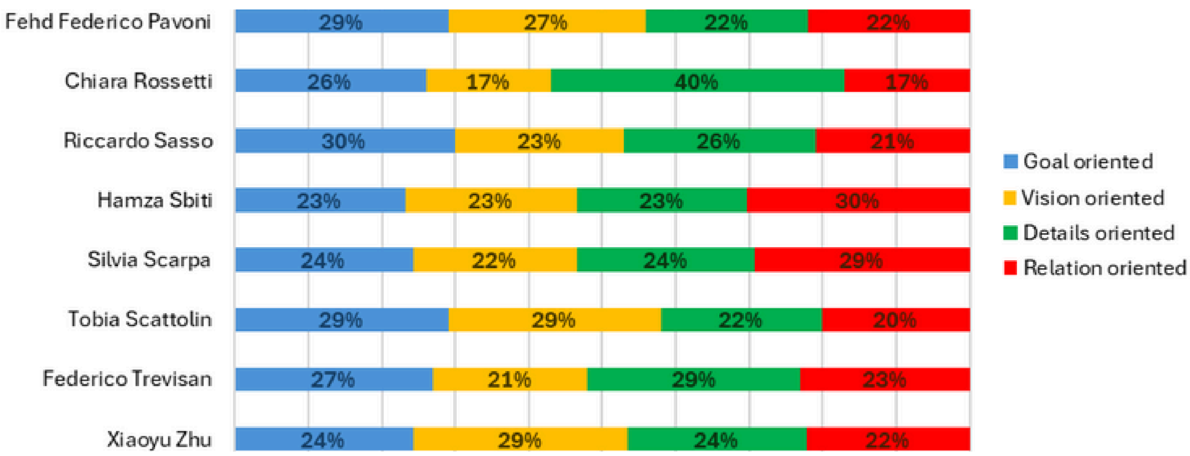
04. Leadership Dynamics

Team Dynamics 40

Personal Growth 44

Team Dynamics

The Esoteric Leaders is a very **heterogeneous** team in the matter of nationality, academic background and individual characteristics: as can be seen in the **Whole Brain**'s results graph, all components have different inclination from each other, which gave the team a complete visual of the group work.



Thanks to the more vision oriented members we were able to **explore** new ideas and develop alternative insights. On the other side, the details oriented ones contributed to give **structure** and order to the work, planning meetings and setting rules when needed. The problem-solving capacities and the logical thinking of the members characterised by a goal tendency helped to **manage** time and resources. Finally, the most relation oriented component brought **enthusiasm** that was essential in the most critical moments.

This multiplicity personality resulted in stimulating factors in terms of exchange of ideas, prospective and personal growth: we gained insights into different cultures, ways of thinking, and approaches to work, leading us to more creative solutions and **innovative thinking**.

On the other hand, it was challenging because of some communication problems and different ways to approach situations.

Due to the fact that the teams were made by the professors, the majority of us didn’t know each other so we dedicated time to team building even though organising meetings was not easy since each member had a different academic schedule.

Together we faced two projects, the Leadership Story and the Innovation Project, that followed very different paths regarding the team dynamics.

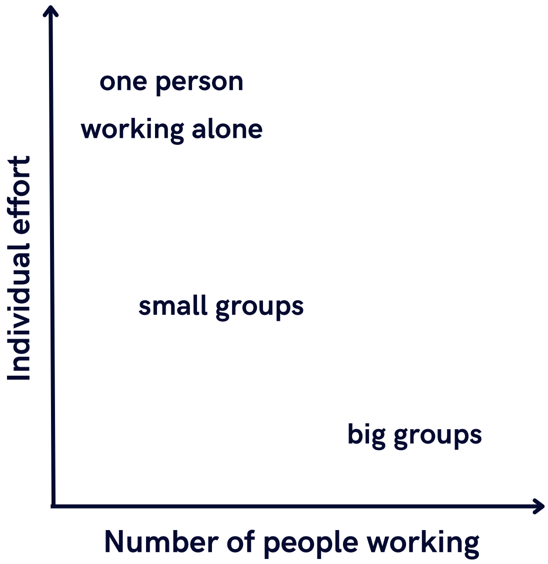
Leadership Story

The Leadership Story has been the starting point of our journey, full of excitement and willingness to face the challenge. Everyone found the assignment very interesting and for this reason, despite the busy schedules and the difficulty in organising physical meetings, there has been **commitment** and **participation** by each member.

We reached a good level of communication through online reviews and briefings, also the workload has been splitted in a proper way: even if some of us managed the contacts with the interviewed, others handled the content part and, finally, some others the design one, the constant progress of the project was clear to everyone leading to a deep satisfaction of the work submitted.

Innovation Project

The Innovation Project followed a completely different path.
Given the previous performance of the team, the expectations were to continue to work in an increasingly structured and smooth direction. But this didn't happen.



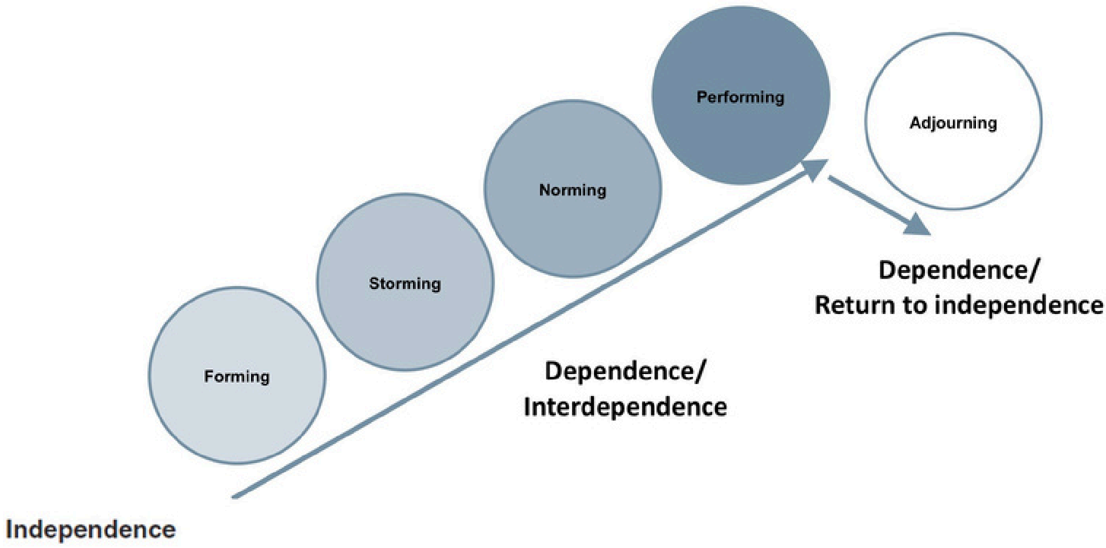
SOCIAL LOAFING
The more people working on a group activity, the less each member will contribute to the group.

In the beginning the actual workload has been underestimated partially because of the incoming deadlines of other courses projects and also the guidelines weren't really clear. The first meetings have been chaotic and unproductive, a general confusion and uncertainty about the requirements had been disheartening.
In addition the team fell into the **social loafing trap**: the awareness to be a numerous group led some of us to diminish the contribution knowing others would compensate.

With the deadline incoming a breakthrough was reached thanks to some members that took over the situation, giving precise instructions, setting intermediate milestones and scheduling regular meetings to keep everything under control.
Thanks to that the team has returned **performing** at the best of its capacity, managing to deliver the work in time with an overall satisfaction of the submission.
In the end we resulted in eight people working on the same project while we should have been more like eight people committed together to one final objective.

Team Dynamics - Conclusions

Looking at the **Tuckman Model**, it can be said that the team managed to reach the performing stage without ever really going through the norming one: we were never able to set a common goal or a shared sense of unity and cooperation which makes a group of people an effective team.



Nevertheless none of us ever felt unsafe or embarrassed to speak up: we succeeded in creating a **safe environment** where propose ideas and explore them critically, take risks and be vulnerable; **conflicts** have occurred, but always overcame through discussions handled with a collaborative mindset which let us to understand the view of others, gain insight of different perspective and learn from ours teammates.

Personal Growth

Fehd Federico Pavoni

BCs Management Engineering - Politecnico di Milano

Working in a group with people you don't know can always be a challenge, especially for someone like me who tends to be introverted. Moreover, everyone has different goals and ways of doing things, so sometimes it can be difficult to align and be efficient, especially because being such in a numerous group could make it hard to get organised.

Fortunately, the group members all showed interest in achieving an excellent result right from the start, and from my point of view, it was very interesting to work with people from abroad who have different cultures and habits from ours.

This project allowed me to develop new skills, particularly better relational and organisational abilities, which I believe are essential for actively participating in the group and creating synergies with other members.

Chiara Rossetti

BCs Management Engineering - Politecnico di Milano

During my academic experience I had a consistent number of team projects and over the years I learnt how to deal with them in the best possible way. Clearly working with new people is always a challenge and the "getting to know each other" part is vital to setting goals and understanding how different members approach teamwork.

At first, the situation overwhelmed me but as time went by we grew as a group.

At the end of this experience I can confidently say that the team dynamics we faced helped me to go out of my comfort zone: I learnt to take control of the situation when needed and to delegate. These two things are not always easy for me for the main reasons that I don't like telling people what to do and, sometimes, I mistakenly think that I can do all by myself. But when the workload is that substantial you can't, and you have to trust the fact that, in the end, you all have a shared objective.

In conclusion I'm thankful to the team for teaching me the importance of collaboration and trust because "*no man is an island*".

Riccardo Sasso

BCs Management Engineering - Università degli Studi di Padova

I would start by saying that it was my first time in a group project, and initially, I was somewhat sceptical about it. In the past, I had jobs where the workload was lighter and could be easily managed by an individual. I'll be honest: in the first leadership assignment, I hadn't realised it yet, but when the innovation one was launched, I understood the importance of working in a team.

The workload was substantial, and a single person wouldn't have been able to handle everything. Coordination and division of tasks were essential to achieve a good result quickly. Being my first time I took a step back and left the room for others to act. Overall, I got along well with the team members, having the opportunity to get to know new thoughts and perspectives. The tension in the air was not absent, but personally, it was handled in the best way. Would I recommend this experience? Absolutely!

Hamza Sbiti

BCs Management Engineering - Ecole Centrale Casablanca

Having extensive experience in working with groups, I always find the dynamic of team projects to be highly enjoyable and stimulating. Previously, I engaged in two innovation projects at my home university, Centrale, which deepened my appreciation for collaborative efforts. The process of bringing together diverse ideas and converging them into a cohesive plan is particularly rewarding.

In this project, what stood out to me was the variation in work styles influenced by cultural backgrounds. Adapting to these differences was a new and enlightening experience. It underscored the importance of flexibility and cultural sensitivity in achieving effective collaboration. This experience not only reinforced my enthusiasm for teamwork, but also enriched my understanding of how adaptability can enhance the overall success of a group effort.

Silvia Scarpa

BCs Management Engineering - Università degli Studi di Bologna

I must confess, I've never been a big fan of group works. I am a sociable person and I like being with other people but my busy schedule often makes me think that working alone is the best way to manage my time and responsibilities effectively.
However, my perspective changed deeply during our first group project: I realised the profound value of diversity within our team and the unique perspective everyone could bring to the work.
Even when we faced the second assignment, more challenging than the first, I learned to trust the contributions of each group member, to work under pressure and I appreciated the spirit of collaboration and mutual support among us.
In conclusion, overcoming challenges together and engaging with new people proved to be immensely rewarding, also thanks to the positive atmosphere we collectively cultivated within our group.

Tobia Scattolin

BCs Management Engineering - Politecnico di Milano

Working on a university project with a group of eight international students was a highly enriching experience. Initially, I was apprehensive about the potential challenges of language barriers and different working styles. However, these diverse perspectives proved to be our greatest asset. Our first meetings were dedicated to understanding each other's strengths and setting clear goals.
Tasks were divided based on individual expertise, ensuring efficiency and allowing everyone to contribute meaningfully.
Despite some initial hurdles, such as coordinating schedules and aligning on expectations, our collaborative spirit prevailed. The experience taught me the importance of flexibility, cultural awareness, and effective communication in a team setting. Overall, this project not only enhanced my academic skills but also broadened my understanding of teamwork in a multicultural environment.

Federico Trevisan

BCs Management Engineering - Politecnico di Milano

Although it was not my first time working in a group, it was the first time I worked in such a large team and with several people I did not already know. Consequently, when the project began, I was enthusiastic and keen to take on the new challenge. The extensive workload highlighted the importance of effective delegation and division of tasks among team members.
It underscored the need to trust others' contributions while ensuring that the final output remains consistent and of high quality. Ultimately, the experience was inspiring. Transitioning from periods of exceptional teamwork to moments of difficulty taught me how to navigate various situations, providing valuable lessons for my future projects.

Xiaoyu Zhu

BCs Financial Management - Donghua Univerity

At first, I was not a big fan of teamwork, as I had learnt from many past experiences that it is usually inefficient and difficult to find teammates who are compatible with each other. In a foreign country, the differences in routine, culture, philosophy and work habits only add to the frustration of working together. But the course is interesting, so why not give teamwork another chance? So I chose to do it again.
Fortunately, my teammates were friendly. And during the previous sessions I realised that we have similar perceptions, which made understanding each other possible. This time I chose to listen more to friends from their country to understand what they were used to than to be a leader. There is no doubt that this cooperation allowed me to deepen my understanding of teamwork, culture and gain more experience.

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